Solution Requirements

Date	26 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Step 1: Functional Requirements (FR)

S.No	**Requirement Category**	**Requirement Description**	**Priority (High/Medium/Low)**
1	Authentication	No authentication is required in the current prototype. Future versions can include administrator login for secure access.	Medium
2	Authorization levels	All users can access the prediction module. Future enhancement: Admins can manage datasets and retrain the model, while users can only make predictions.	Medium
3	External interfaces	The system interacts with a trained Machine Learning model (best_model.pkl) and the saved StandardScaler	High

		(scaler.pkl) through the Flask backend. Future versions can integrate weather APIs for real-time data.	
4	Transactions processing	The system accepts weather parameters, preprocesses them, performs flood prediction using the trained model, and returns the prediction result without storing user input.	High
5	Reporting	Displays prediction results ("Flood Prediction, Flood Probability, and Risk Level"). Future versions can include prediction reports and dashboards.	Medium
6	Business rules, etc.	All input values must be numeric and validated before prediction. The model uses the same feature scaling as during training, and only the trained model is used for inference.	High
7	Compliance to laws or regulations	The application does not collect sensitive personal information. Future deployments should comply with applicable data privacy regulations and cybersecurity best practices.	Low

8	*Other*	The application should provide quick, accurate flood predictions through a simple web interface to support disaster preparedness and decision-making.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application should process prediction requests quickly.	Prediction displayed within **2 seconds**.
2	Scalability	The system should support multiple prediction requests and allow future cloud deployment.	Designed for future scalability and cloud deployment.
3	Security & Data Privacy	Validate all user inputs and prevent invalid or malicious data submission.	Input validation implemented; no sensitive user data stored.

4	Reliability & Availability	The application should provide consistent and reliable flood predictions during normal operation.	Reliable execution with consistent prediction results during local operation.
5	Usability & Accessibility	The interface should be simple, responsive, and easy for disaster management authorities and other users to operate.	Users can complete a prediction in **less than 5 sec** with a responsive UI.
6	*Other*	Maintainability and portability. The code should be modular, well-documented, and deployable on cloud platforms such as IBM Cloud or Render.	Modular codebase with reusable components and cloud deployment support.